

Project Design Phase
Solution Architecture

Date	03 October 2026
Team ID	SWTID-2026-4647
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

The **Auto Ticket Classification System** is built on the **ServiceNow platform**, designed to automate IT helpdesk ticket categorization using a no-code approach. The architecture ensures seamless data flow, scalability, and real-time automation.

1. User Interface Layer

- **Actors:** Students and teachers.
- **Function:** Submit IT support tickets through the ServiceNow portal by entering their name and short description.
- **Outcome:** Ticket creation triggers the automation workflow.

2. Application Logic Layer

- **Component:** ServiceNow Flow Designer.
- **Function:** Automatically analyzes the short description for keywords such as *Wi-Fi*, *Projector*, *Password*, or *Slow*.
- **Process:**
 - Wi-Fi → Category: Network, Subcategory: Wi-Fi
 - Projector → Category: Hardware, Subcategory: Projector
 - Password/Login → Category: Access, Subcategory: Forgot Password
 - Slow/Hanging → Category: Performance, Subcategory: Slow Computer
- **Outcome:** Category and Subcategory are assigned automatically.

3. Data Layer

- **Component:** Custom *Incident Workflow* table.
- **Function:** Stores structured ticket data including Caller, Category, Subcategory, State, Assigned Group, and Assigned To.
- **Outcome:** Maintains consistent and retrievable records for IT staff.

4. Notification Layer

- **Component:** ServiceNow Notification API.
- **Function:** Sends automated email confirmation to the caller after ticket creation.

- **Outcome:** Improves transparency and user satisfaction.

5. Infrastructure Layer

- **Component:** ServiceNow Cloud Platform.
- **Function:** Provides secure, scalable, and always-available hosting for the application.
- **Outcome:** Ensures reliability, performance, and accessibility across devices.

Solution Architecture Diagram:

